



UNIVERSITAS AIRLANGGA

Terakreditasi Unggul/Accredited Excellent: SK BAN PT No. 1017/SK/BAN-PT/Ak.Ppj/PT/XI/2023
Kode Perguruan Tinggi/University code number: 001004

dengan ini menyatakan bahwa/*this is to certify that:*

Ratu Noor Hasannah

Nomor Induk Mahasiswa *Student identification Number*: 162112133112

Nomor Induk Kependudukan/*National ID Number*: 3275045509030006

Lahir di: Bekasi, 15 September 2003
born in: Bekasi, 15 September 2003

telah menyelesaikan pendidikan dengan baik pada/*has satisfactorily completed the studies in the:*
PROGRAM SARJANA PROGRAM STUDI TEKNOLOGI SAINS DATA (49202)
Bachelor Program in Data Science Technology Study Program (49202)

FAKULTAS TEKNOLOGI MAJU DAN MULTIDISIPLIN
Faculty of Advanced Technology and Multidiscipline

Tanggal Lulus/*Graduation date*: 25 Juli 2025

Kepadanya diberikan ijazah dan gelar/*has been awarded the certificate and degree of:*

SARJANA SAINS DATA (S.Si.D)
BACHELOR OF SCIENCE

beserta segala hak dan kewajiban yang melekat pada gelar tersebut
With all the rights and responsibilities appertaining thereunto

DEKAN/Dean

Prof. Dr. Ir. Retna Apsari, M.Si., IPM.
NIP 196806261993032003



Diterbitkan di Surabaya, 12 September 2025
Issued in

REKTOR/Rector

Prof. Dr. M. Madyan, S.E., M.Si., M.Fin.
NIP 197101041998021001



TRANSKRIP AKADEMIK

ACADEMIC TRANSCRIPT

Nomor : 162112133112/001004/16.2/S1/2025

PROGRAM STUDI: TEKNOLOGI SAINS DATA

STUDY PROGRAM: DATA SCIENCE TECHNOLOGY



Nama / Name : Ratu Noor Hasannah
NIM / Registration Number : 162112133112
Tempat, tanggal lahir /
Place, Date of Birth : Bekasi, 15 September 2003 / Bekasi, 15
September 2003
Akreditasi Institusi /
Institution Accreditation : A
SK Akreditasi Institusi /
Institution Accreditation
Number :
Asal Sekolah / Previous
School : SMAN 5 TAMBUN SELATAN

Tanggal Terdaftar /
Registration date : 05 September 2021 / 05 September 2021
Tanggal Lulus / Graduation :
date : 25 Juli 2025 / 25 July 2025
Nomor Ijazah / Certificate :
Number :
Gelar Akademik /
Academic Degree : SARJANA SAINS DATA (S.Si.D) / Bachelor
of Science
sks / total credit : 148
IPK / GPA : 3.44

Judul Skripsi : Implementasi Kombinasi Kalman Filter dan LSTM pada Prediksi Harga Minyak Mentah West Texas Intermediate (WTI)
Thesis Title : Implementation Of Kalman Filter And LSTM Combination In Predicting The West Texas Intermediate (WTI) Crude Oil Prices

Kode Code	Mata Kuliah / Subject	sks crd	Nilai Grade
AGI101	Agama Islam I / Religion Islamic I	2	A
AGI401	Agama Islam II / Islamic Studies II	2	A
SIA109	Algoritma Pemrograman I / Programming Algorithm I	3	B
SIA206	Algoritma Pemrograman II / Programming Algorithm II	3	AB
MAS247	Analisis Data Medis / Medical Data Analysis	3	AB
MAS317	Analisis Data Spasial / Analysis of Spasial Data	3	A
MAS330	Analisis Ketahanan Hidup / Survival Analysis	3	A
BAI101	Bahasa Indonesia / Indonesian Language	2	A
BAE110	Bahasa Inggris / English	2	A
BAE312	Bahasa Inggris II / English II	2	AB
SID201	Basis Data / Database	4	A
BID101	Biologi Dasar / General Biology	2	BC
SID207	Data Mining I / Data Mining I	4	AB
SID306	Data Mining II / Data Mining II	4	BC
SIP107	Data dan Pustaka / Data and Literature	2	A
SID205	Eksplorasi dan Visualisasi Data / Data Exploration and Visualization	3	AB
FID113	Fisika Dasar / Basic Physics	2	B
KNJ401	KKN-BBM / Community Services	3	A
PNJ301	Keahlian Riset dan Pengembangan Akademik / Research and Academic Development Skills	3	AB
NOP104	Kewarganegaraan / Civic Education	2	A
MNW201	Kewirausahaan / Entrepreneurship	2	AB
KID105	Kimia Dasar / Fundamental of Chemistry	2	BC
MAS246	Komputasi Statistika / Statistical Computing	3	B
MNM106	Komunikasi dan Pengembangan Diri / Communication and Self Development	2	A
PHP103	Logika dan Pemikiran Kritis / Logic and Critical Thinking	2	A
SJ303	Manajemen Jaringan / Network Management	3	A
MAT108	Matematika / Mathematics	3	BC

Kode Code	Mata Kuliah / Subject	sks crd	Nilai Grade
MAL204	Matematika Diskrit / Discrete Mathematics	3	A
MAT110	Matematika Lanjut / Advanced Mathematics	3	BC
MAA208	Metode Matematika / Mathematical Method	3	A
PNJ302	Metode Penelitian / Research Methods	2	B
MAS104	Metode Statistika / Statistical Methods	4	AB
MAS328	Multivariat / Multivariat	3	B
SIC304	Natural Language Processing / Natural Language Processing	3	B
MAS334	Official Statistics / Official Statistics	3	AB
NOP103	Pancasila / Five Principles of Indonesia	2	A
MAS422	Pemodelan Data Eksperimen / Experimental Data Modeling	3	AB
MAS332	Pemodelan Runtun Waktu / Time Series Modeling	3	AB
SID206	Pengantar Basis Data / Introduction to Database	4	AB
SIC203	Pengantar Kecerdasan Buatan / Introduction to Artificial Intelligence	3	AB
MNM107	Pengantar Kolaborasi Keilmuan / Introduction to Collaborative Science	2	A
SIA104	Pengantar Pemrograman / Introduction to Programming	2	AB
MAS120	Pengantar Statistika / Introduction to Statistics	3	A
TNM101	Pengantar Teknologi Maju / Introduction to Advanced Technology	2	A
KLJ301	Praktek Kerja Lapangan / Industrial Experiences	2	A
MAS106	Probabilitas / Probability	4	B
PNJ495	Proposal Skripsi / Thesis Proposal	2	AB
MAS223	Proses Stokastik / Stochastic Proces	3	B
MAS212	Riset Operasi / Operations Research	3	B
MAS331	Sains Data Consulting / Sains Data Consulting	3	AB
PNJ499	Skripsi / Thesis	4	AB
MAS316	Statistika Matematik / Mathematical Statistics	3	C
MAS245	Statistika Nonparametrik / Nonparametric Statistics	3	B
TNM102	Teknologi Hijau / Green Technology	2	B



TRANSKRIP AKADEMIK

ACADEMIC TRANSCRIPT

Nomor : 162112133112/001004/16.2/S1/2025

PROGRAM STUDI: TEKNOLOGI SAINS DATA

STUDY PROGRAM: DATA SCIENCE TECHNOLOGY



Nama / Name : Ratu Noor Hasannah
NIM / Registration Number : 162112133112
Tempat, tanggal lahir /
Place, Date of Birth : Bekasi, 15 September 2003 / Bekasi, 15
September 2003
Akreditasi Institusi /
Institution Accreditation : A
SK Akreditasi Institusi /
Institution Accreditation
Number :
Asal Sekolah / Previous
School : SMAN 5 TAMBUN SELATAN

Tanggal Terdaftar / : 05 September 2021 / 05 September 2021
Registration date
Tanggal Lulus / Graduation : 25 Juli 2025 / 25 July 2025
date
Nomor Ijazah / Certificate :
Number
Gelar Akademik / : SARJANA SAINS DATA (S.Si.D) / Bachelor
Academic Degree of Science
sks / total credit : 148
IPK / GPA : 3.44

Judul Skripsi : Implementasi Kombinasi Kalman Filter dan LSTM pada Prediksi Harga Minyak Mentah West Texas Intermediate (WTI)
Thesis Title : Implementation Of Kalman Filter And LSTM Combination In Predicting The West Texas Intermediate (WTI) Crude Oil Prices

Kode Code	Mata Kuliah / Subject	sks crd	Nilai Grade
AGI101	Agama Islam I / Religion Islamic I	2	A
AGI401	Agama Islam II / Islamic Studies II	2	A
SIA109	Algoritma Pemrograman I / Programming Algorithm I	3	B
SIA206	Algoritma Pemrograman II / Programming Algorithm II	3	AB
MAS247	Analisis Data Medis / Medical Data Analysis	3	AB
MAS317	Analisis Data Spasial / Analysis of Spasial Data	3	A
MAS330	Analisis Ketahanan Hidup / Survival Analysis	3	A
BAI101	Bahasa Indonesia / Indonesian Language	2	A
BAE110	Bahasa Inggris / English	2	A
BAE312	Bahasa Inggris II / English II	2	AB
SID201	Basis Data / Database	4	A
BID101	Biologi Dasar / General Biology	2	BC
SID207	Data Mining I / Data Mining I	4	AB
SID306	Data Mining II / Data Mining II	4	BC
SIP107	Data dan Pustaka / Data and Literature	2	A
SID205	Eksplorasi dan Visualisasi Data / Data Exploration and Visualization	3	AB
FID113	Fisika Dasar / Basic Physics	2	B
KNJ401	KKN-BBM / Community Services	3	A
PNJ301	Keahlian Riset dan Pengembangan Akademik / Research and Academic Development Skills	3	AB
NOP104	Kewarganegaraan / Civic Education	2	A
MNW201	Kewirausahaan / Entrepreneurship	2	AB
KID105	Kimia Dasar / Fundamental of Chemistry	2	BC
MAS246	Komputasi Statistika / Statistical Computing	3	B
MNM106	Komunikasi dan Pengembangan Diri / Communication and Self Development	2	A
PHP103	Logika dan Pemikiran Kritis / Logic and Critical Thinking	2	A
SJ303	Manajemen Jaringan / Network Management	3	A
MAT108	Matematika / Mathematics	3	BC

Kode Code	Mata Kuliah / Subject	sks crd	Nilai Grade
MAL204	Matematika Diskrit / Discrete Mathematics	3	A
MAT110	Matematika Lanjut / Advanced Mathematics	3	BC
MAA208	Metode Matematika / Mathematical Method	3	A
PNJ302	Metode Penelitian / Research Methods	2	B
MAS104	Metode Statistika / Statistical Methods	4	AB
MAS328	Multivariat / Multivariat	3	B
SIC304	Natural Language Processing / Natural Language Processing	3	B
MAS334	Official Statistics / Official Statistics	3	AB
NOP103	Pancasila / Five Principles of Indonesia	2	A
MAS422	Pemodelan Data Eksperimen / Experimental Data Modeling	3	AB
MAS332	Pemodelan Runtun Waktu / Time Series Modeling	3	AB
SID206	Pengantar Basis Data / Introduction to Database	4	AB
SIC203	Pengantar Kecerdasan Buatan / Introduction to Artificial Intelligence	3	AB
MNM107	Pengantar Kolaborasi Keilmuan / Introduction to Collaborative Science	2	A
SIA104	Pengantar Pemrograman / Introduction to Programming	2	AB
MAS120	Pengantar Statistika / Introduction to Statistics	3	A
TNM101	Pengantar Teknologi Maju / Introduction to Advanced Technology	2	A
KLJ301	Praktek Kerja Lapangan / Industrial Experiences	2	A
MAS106	Probabilitas / Probability	4	B
PNJ495	Proposal Skripsi / Thesis Proposal	2	AB
MAS223	Proses Stokastik / Stochastic Procees	3	B
MAS212	Riset Operasi / Operations Research	3	B
MAS331	Sains Data Consulting / Sains Data Consulting	3	AB
PNJ499	Skripsi / Thesis	4	AB
MAS316	Statistika Matematik / Mathematical Statistics	3	C
MAS245	Statistika Nonparametrik / Nonparametric Statistics	3	B
TNM102	Teknologi Hijau / Green Technology	2	B

Keterangan / Description

Nilai Huruf / Value	A	AB	B	BC	C	D	E
Bobot / Quality	4	3.5	3	2.5	2	1	0

Kriteria Kelulusan / Graduation Criteria

IPK / GPA	Predikat / Predicate
2 - 2.75	Lulus / Graduate
2.76 - 3	Memuaskan / Satisfactory
3.01 - 3.5	Sangat Memuaskan / Fully Satisfactory
3.51 - 4	Sangat Memuaskan / Fully Satisfactory

Diterbitkan di Surabaya pada tanggal 04 Agustus 2025

Published in Surabaya, 04 August 2025

Dekan / Dean,

Prof. Dr. Dwi Setyawan, S.Si., MSi., Apt
NIP. 197111301997031003

Catatan : Predikat dengan pujian diberikan dengan memperhatikan masa studi maksimum 4 (Empat) tahun.
Note : Graduation predicate of With Honors is given by observing the maximum study period of 4 (Four) years